

E-Learning Platform Usability Analysis

A Statistical Usability Evaluation using Hypothesis Testing and UX Metrics

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1 Introduction

Usability evaluation is a critical aspect of user experience (UX) design, particularly for elearning platforms where efficient interaction and positive learner perception directly influence learning outcomes. An effective course player should allow users to complete tasks with minimal effort while maintaining satisfactory levels of engagement and satisfaction.

This study evaluates the usability of a redesigned e-learning course player developed to improve task efficiency and learner satisfaction. Four layout variants (Designs A–D) were created, each available in three color themes (Light, Mid and Dark) and three base font sizes (Small, Medium and Large). The prototypes were tested across different device types, including desktop, mobile and tablet. During the usability testing sessions, participants completed core tasks such as finding lesson notes, adjusting playback speed and marking lessons as complete.

The aim of this report is to analyze the usability testing data using appropriate statistical methods to assess whether the redesigned course player achieves improvements in task efficiency and learner satisfaction. Two statistical analyses are conducted in this report. First, a paired-sample t-test is used to examine whether there is a significant difference in task completion time between two layout designs (Design A and Design B) for the same users, addressing task efficiency. Second, Pearson's correlation coefficient is used to investigate the relationship between interaction effort, measured by the number of clicks for Design A, and overall learner satisfaction reported during the usability testing session. The outcomes of these analyses are used to draw evidence-based conclusions and propose usability improvements for the course player interface.

2 Hypothesis testing

In this report, I will discuss a study that examines:

1. How the layout design of an e-learning course player affects **task efficiency**, measured by task completion time.
2. Whether **learner satisfaction** is related to interaction effort, measured by the number of clicks during task completion.

Results collected to evaluate the formed hypotheses will be analysed using two statistical methods. A paired-sample t-test will be used to determine whether there is a significant difference in task completion time between two layout designs (Design A and Design B) for the same users. In addition, Pearson's correlation coefficient will be used to examine the relationship between interaction effort (number of clicks for Design A) and overall learner satisfaction reported during the usability testing session.

2.2 Paired sample t-test

This section presents the first hypothesis, which focuses on task efficiency for desktop users only. A paired-sample t-test is used to determine whether there is a statistically significant difference in task completion time between two layout designs (Design A and Design B) when evaluated by the same desktop users.

2.2.1 Hypothesis testing for Paired sample t-test

Research question	Does the layout design significantly affect task efficiency (measured by task completion time) for desktop users?
Hypothesis statement	H ₀ : There is no significant difference in task completion time between Design A and Design B for desktop users.
	H _a : There is a significant difference in task completion time between Design A and Design B for desktop users.
Mathematical expression of the Hypothesis	H₀: $\mu_A = \mu_B$ μ_A = mean task completion time for Design A (desktop users) μ_B = mean task completion time for Design B (desktop users)
	H _a : $\mu_A \neq \mu_B$
Dependent variable	Task completion time
Dependent variable data type	Ratio
Independent variable	Layout design (Design A vs Design B)
Independent variable type	Nominal
Control variable	Device type (Desktop only), Color theme, Font size, Task type
Experimental design	within-subjects
Study method	Paired-sample t-test

2.2.2 Data organisation

For the paired-sample t-test analysis, only data from desktop users were selected from the dataset in order to control for the effect of device type. The analysis focuses on task efficiency, measured by task completion time, for two layout designs (Design A and Design B).

The following columns were extracted from the dataset for this analysis:

- Participant ID
- Task Completion Time for Design A
- Task Completion Time for Design B

Data preparation

The raw dataset was imported into SPSS and organized to support the paired-sample t-test. The data were filtered to include desktop users only, and only the relevant columns required for the analysis were selected.

The following variables were used:

- Task completion time for Design A
- Task completion time for Design B

Each row represents one desktop participant, with task completion times for Design A and Design B recorded in separate columns so that results from the same participant could be compared.

The dataset was checked for invalid entries, and any invalid values were removed before conducting the analysis. Screenshots of the SPSS Variable View and Data View are provided to show how the data were organized.

Variable view

	Name	Type	Width	Decimals	Label	Values	Missing	Columns	Align	Measure	Role
1	participants	Numeric	6	0	Participant ID	None	None	12	Right	Scale	Input
2	Time_A	Numeric	6	0	Task completio...	None	None	12	Right	Scale	Input
3	Time_B	Numeric	6	0	Task completio...	None	None	12	Right	Scale	Input
4											

Data view

	participants	Time_A	Time_B
1	1	72	97
2	2	91	104
3	3	83	101
4	4	80	103
5	5	80	98
6	6	84	99
7	7	75	101
8	8	75	97
9	9	81	101
10	10	79	105
11	11	83	101
12	12	87	93
13	14	79	90
14	15	84	101
15	16	80	103
16	17	77	103
17	19	77	104
18	20	80	100
19	21	90	100
20	22	86	102
21	23	83	96
22	24	81	94
23	25	85	98
24	26	73	102
25	27	88	97
26	28	82	98
27	29	79	98
28	30	82	98
29	31	89	102
30	33	74	105
31	34	80	96
32	35	75	100
33	37	81	99
34	38	86	98
35	39	79	100
36	40	84	100

Data organizing

Case Processing Summary

	Valid		Cases Missing		Total	
	N	Percent	N	Percent	N	Percent
Task completion time for Design A (Desktop users)	36	100.0%	0	0.0%	36	100.0%
Task completion time for Design B (Desktop users)	36	100.0%	0	0.0%	36	100.0%

Frequencies

		Statistics	
		Task completion time for Design A (Desktop users)	Task completion time for Design B (Desktop users)
N	Valid	36	36
	Missing	0	0
Mean		81.22	99.56
Median		81.00	100.00
Mode		80	98
Std. Deviation		4.752	3.333
Variance		22.578	11.111
Skewness		.084	-.660
Std. Error of Skewness		.393	.393
Range		19	15
Minimum		72	90
Maximum		91	105
Sum		2924	3584
Percentiles	25	79.00	98.00
	50	81.00	100.00
	75	84.00	102.00

Frequency Table

Task completion time for Design A (Desktop users)					
		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	72	1	2.8	2.8	2.8
	73	1	2.8	2.8	5.6
	74	1	2.8	2.8	8.3
	75	3	8.3	8.3	16.7
	77	2	5.6	5.6	22.2
	79	4	11.1	11.1	33.3
	80	5	13.9	13.9	47.2
	81	3	8.3	8.3	55.6
	82	2	5.6	5.6	61.1
	83	3	8.3	8.3	69.4
	84	3	8.3	8.3	77.8
	85	1	2.8	2.8	80.6
	86	2	5.6	5.6	86.1
	87	1	2.8	2.8	88.9
	88	1	2.8	2.8	91.7
	89	1	2.8	2.8	94.4
	90	1	2.8	2.8	97.2
	91	1	2.8	2.8	100.0
Total		36	100.0	100.0	

Task completion time for Design B (Desktop users)

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	90	1	2.8	2.8	2.8
	93	1	2.8	2.8	5.6
	94	1	2.8	2.8	8.3
	96	2	5.6	5.6	13.9
	97	3	8.3	8.3	22.2
	98	6	16.7	16.7	38.9
	99	2	5.6	5.6	44.4
	100	5	13.9	13.9	58.3
	101	5	13.9	13.9	72.2
	102	3	8.3	8.3	80.6
	103	3	8.3	8.3	88.9
	104	2	5.6	5.6	94.4
	105	2	5.6	5.6	100.0
	Total	36	100.0	100.0	

2.2.3 Descriptive statistics and discussion of results

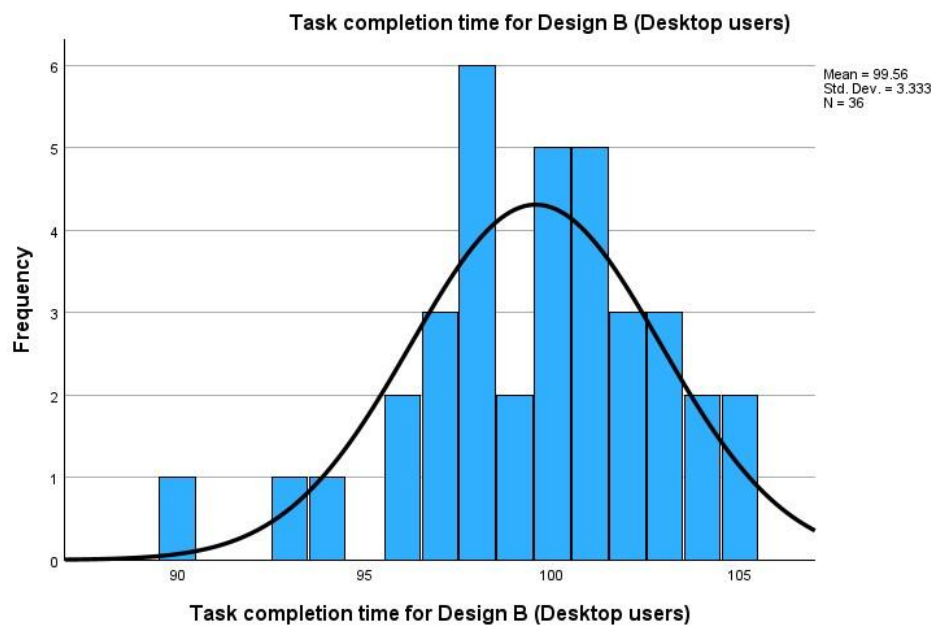
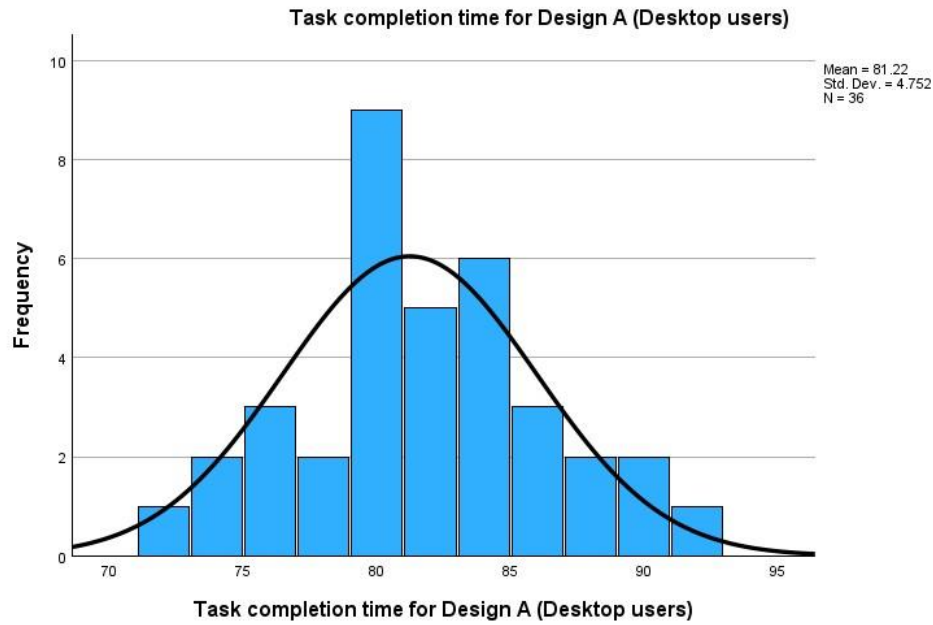
Descriptive statistics were calculated for task completion time for Design A and Design B using data from desktop users. Measures of central tendency and variability were used to summarise and compare the task efficiency of the two designs

Design	Measures of Central Tendency			Measures of Variability			
	Mean	Median	Mode	Range	Variance	Standard Deviation	Skewness
A	81.22	81.00	80	19	22.578	4.752	0.084
B	99.56	100.00	98	15	11.111	3.333	-0.660

Graphical Representation

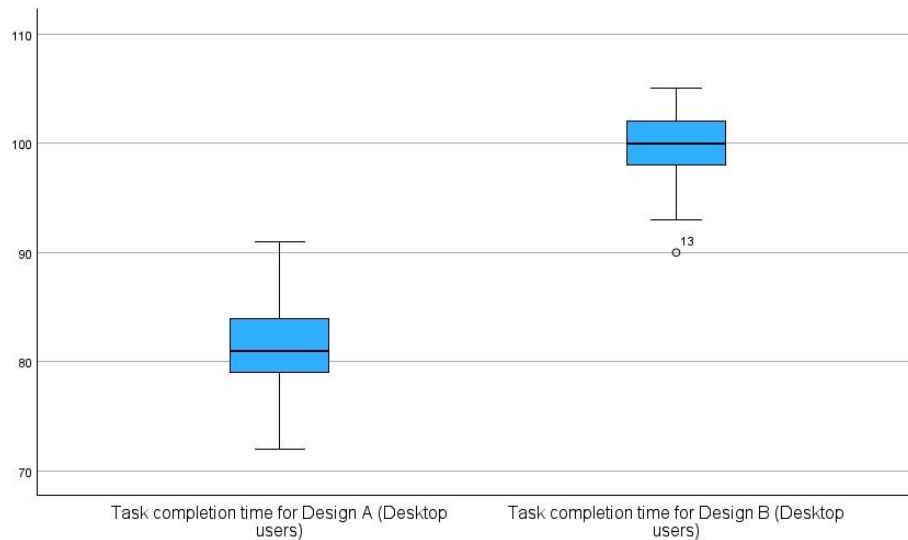
Histograms

A histogram was selected to represent task completion time because it effectively shows the shape and spread of the data, supporting the assessment of distribution characteristics and normality.



Box Plot

A box plot was used to compare task completion time between Design A and Design B, as it clearly displays the median, spread and potential outliers, enabling a direct comparison of task efficiency between the two designs.



Analysis of Descriptive Statistics

For desktop users, Design A shows better task efficiency than Design B. The mean task completion time for Design A (81.22) is lower than for Design B (99.56), indicating that users completed tasks faster using Design A. The median and mode values also support this finding, as both are lower for Design A compared to Design B.

In terms of variability, Design A shows greater variation in task completion time than Design B, as reflected by its higher standard deviation and range. The skewness values for both designs are close to zero, indicating that task completion times for both designs are approximately normally distributed.

Analysis of Graphical Representations

Histogram Analysis

The histogram for Design A appears relatively symmetric, indicating a near-normal distribution of task completion times. The histogram for Design B also shows an approximately normal distribution, with a slight skew. These observations are consistent with the skewness values and support the assumption of normality for both designs.

Box Plot Analysis

The box plot shows that Design A has a lower median task completion time compared to Design B, confirming that tasks are completed faster using Design A. Design A also displays a wider spread of values, indicating greater variability in user performance. In contrast, Design B shows a higher median with a narrower spread, suggesting slower but more consistent task completion times.

Summary

The normality of the data was assessed using both descriptive statistics and graphical representations. The skewness values for task completion time for Design A (0.084) and Design B (-0.660) are close to zero, indicating that both datasets are approximately normally distributed. This assessment is supported by the histograms, which show roughly symmetric distributions and by the box plot, which shows no extreme deviations.

As the assumption of normality is satisfied, it is appropriate to proceed with parametric statistical analysis to examine the significance of the results. Therefore, a paired-sample t-test can be used for this hypothesis.

If the data had not met the normality assumption, the appropriate non-parametric alternative would have been the Wilcoxon signed-rank test.

2.2.4 Conducting Paired sample t-test, interpret and report the results

Method Recap

A paired-sample t-test was conducted to examine whether there was a statistically significant difference in task completion time between Design A and Design B for desktop users. This test was appropriate as the same participants completed tasks using both designs.

Descriptive Recap

The mean task completion time for,

- Design A = 81.22 seconds (SD = 4.75)
- Design B = 99.56 seconds (SD = 3.33)

This indicates that, on average, users completed tasks faster when using Design A compared to Design B.

Significance Test Results

The paired-sample t-test revealed a statistically significant difference in task completion time between Design A and Design B,

- $t(35) = -18.26$
- $p < .001$ (two-tailed)

The mean difference in task completion time was -18.33 seconds, with a 95% confidence interval ranging from -20.37 to -16.30 seconds.

Hypothesis Test Recap

Null Hypothesis (H_0): There is no significant difference in task completion time between Design A and Design B for desktop users.

Alternative Hypothesis (H_a): There is a significant difference in task completion time between Design A and Design B for desktop users.

As the p-value is less than the significance level of 0.05, the null hypothesis is rejected. This result indicates that layout design has a significant effect on task efficiency for desktop users, with Design A demonstrating significantly better performance than Design B.

Reliability of the Analysis

The descriptive statistics and graphical analyses confirmed that task completion time for both designs was approximately normally distributed. Therefore, the assumptions required for conducting a paired-sample t-test were satisfied and the results of this analysis are considered reliable.

Paired Samples Statistics

		Mean	N	Std. Deviation	Std. Error Mean
Pair 1	Task completion time for Design A (Desktop users)	81.22	36	4.752	.792
	Task completion time for Design B (Desktop users)	99.56	36	3.333	.556

Paired Samples Correlations

		N	Correlation	Significance	
				One-Sided p	Two-Sided p
Pair 1	Task completion time for Design A (Desktop users) & Task completion time for Design B (Desktop users)	36	-.082	.317	.635

Paired Samples Test

		Paired Differences					Significance		
		Mean	Std. Deviation	Std. Error Mean	95% Confidence Interval of the Difference		t	df	
					Lower	Upper			One-Sided p
Pair 1	Task completion time for Design A (Desktop users) - Task completion time for Design B (Desktop users)	-18.333	6.024	1.004	-20.371	-16.295	-18.261	35	<.001
									<.001

Paired Samples Effect Sizes

					95% Confidence Interval	
			Standardizer ^a	Point Estimate	Lower	Upper
Pair 1	Task completion time for Design A (Desktop users) - Task completion time for Design B (Desktop users)	Cohen's d	6.024	-3.044	-3.821	-2.258
		Hedges' correction	6.157	-2.978	-3.738	-2.209

a. The denominator used in estimating the effect sizes.

Cohen's d uses the sample standard deviation of the mean difference.

Hedges' correction uses the sample standard deviation of the mean difference, plus a correction factor.

2.4 Pearson's Correlation Coefficient

This section examines whether there is a relationship between interaction effort and learner satisfaction. Interaction effort is measured by the number of clicks for Design A, while learner satisfaction is measured using the overall satisfaction level reported during the usability testing session.

2.4.1 Hypothesis testing for Independent sample t-test

Research question	Is there a relationship between interaction effort (measured by the number of clicks for Design A) and overall learner satisfaction?
Hypothesis statement	H_0 : There is no significant relationship between the number of clicks for Design A and overall learner satisfaction.
	H_a : There is a significant relationship between the number of clicks for Design A and overall learner satisfaction.
Mathematical expression of the Hypothesis	$H_0: r = 0$ $r = \text{Pearson's correlation coefficient}$
	$H_a: r \neq 0$
Dependent variable	learner satisfaction
Dependent variable data type	Interval
Independent variable	Interaction effort (number of clicks for Design A)
Independent variable type	Ratio
Control variable	Device type (desktop users only), layout design (Design A), colour theme and font size
Experimental design	within-subjects
Study method	Pearson correlation

2.4.2 Data organisation

For the Pearson correlation analysis, only data from desktop users were selected from the dataset in order to control for the effect of device type. The analysis focuses on the relationship between interaction effort, measured by the number of clicks for Design A, and overall learner satisfaction reported during the usability testing session.

Data preparation

The raw dataset was imported into SPSS and organized to support Pearson correlation analysis. The data were filtered to include desktop users only and only the relevant columns required for the analysis were selected. The following variables were used:

- Number of clicks for Design A
- Satisfaction level

Each row represents one desktop participant, with interaction effort and satisfaction values recorded in separate columns to allow correlation analysis. The dataset was checked for invalid entries, and any unsuitable values were removed before conducting the analysis. Screenshots of the SPSS Variable View and Data View are provided to show how the data were organized.

Variable view

	Name	Type	Width	Decimals	Label	Values	Missing	Columns	Align	Measure	Role
1	participants	Numeric	5	0	participant ID	None	None	12	Right	Scale	Input
2	Num_error...	Numeric	8	0	Number of clic...	None	None	12	Right	Scale	Input
3	Satisfaction...	Numeric	5	0	Satisfaction lev...	None	None	12	Right	Ordinal	Input

Data view

	 participants	 Num_errors _Design_A	 Satisfaction _level
1	1	152	5
2	2	171	1
3	3	163	2
4	4	160	2
5	6	164	2
6	7	155	5
7	8	155	5
8	9	161	3
9	10	159	3
10	11	163	3
11	12	167	2
12	13	159	3
13	14	159	3
14	15	164	2
15	16	160	3
16	17	157	5
17	18	166	2
18	20	160	3
19	21	170	1
20	22	166	2
21	23	163	2
22	24	161	3
23	25	165	2
24	26	153	5
25	27	168	2
26	28	162	3
27	29	159	3
28	30	162	3
29	31	169	1
30	32	161	3
31	35	155	5
32	36	163	2
33	37	161	2
34	38	166	2
35	39	159	3
36	40	164	2

Data organizing

Frequencies

Statistics			
		Number of clicks Design A	Satisfaction level - 5 point Likert scale: (1) very unsatisfied - (5) very satisfied
N	Valid	36	36
	Missing	0	0
Mean		161.72	2.78
Median		161.50	3.00
Mode		159	2
Std. Deviation		4.596	1.174
Variance		21.121	1.378
Skewness		-.080	.796
Std. Error of Skewness		.393	.393
Range		19	4
Minimum		152	1
Maximum		171	5
Sum		5822	100
Percentiles	25	159.00	2.00
	50	161.50	3.00
	75	164.75	3.00

Frequency Table

Number of clicks Design A					
		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	152	1	2.8	2.8	2.8
	153	1	2.8	2.8	5.6
	155	3	8.3	8.3	13.9
	157	1	2.8	2.8	16.7
	159	5	13.9	13.9	30.6
	160	3	8.3	8.3	38.9
	161	4	11.1	11.1	50.0
	162	2	5.6	5.6	55.6
	163	4	11.1	11.1	66.7
	164	3	8.3	8.3	75.0
	165	1	2.8	2.8	77.8
	166	3	8.3	8.3	86.1
	167	1	2.8	2.8	88.9
	168	1	2.8	2.8	91.7
	169	1	2.8	2.8	94.4
	170	1	2.8	2.8	97.2
	171	1	2.8	2.8	100.0
Total		36	100.0	100.0	

Satisfaction level - 5 point Likert scale: (1) very unsatisfied - (5) very satisfied

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	1	3	8.3	8.3	8.3
	2	14	38.9	38.9	47.2
	3	13	36.1	36.1	83.3
	5	6	16.7	16.7	100.0
	Total	36	100.0	100.0	

2.4.3 Descriptive statistics and discussion of results

Descriptive statistics were calculated to summarize the key characteristics of interaction effort (number of clicks for Design A) and overall learner satisfaction for desktop users. Measures of central tendency and variability were used to provide an overview of the distribution and spread of both variables prior to correlation analysis.

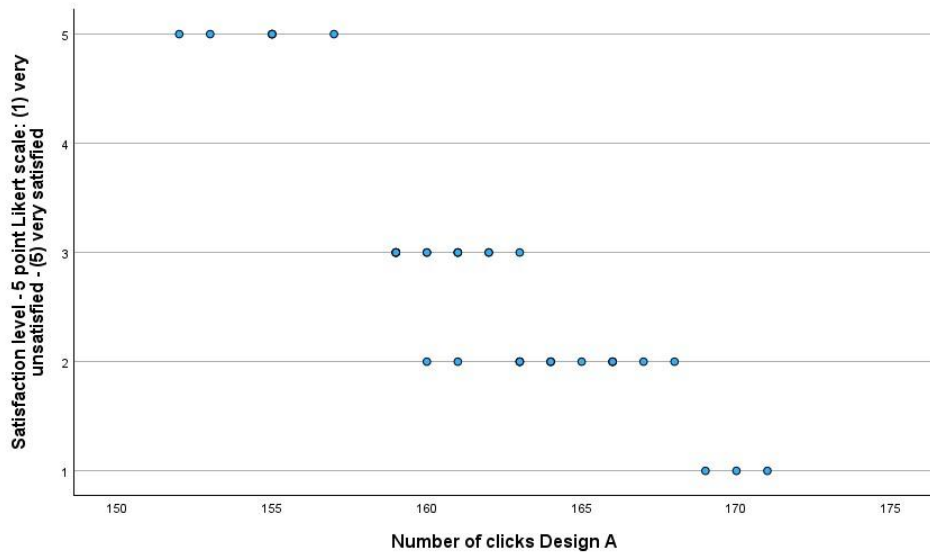
The descriptive results show the average number of clicks required to complete tasks using Design A, as well as the overall satisfaction levels reported by users. Variability measures indicate how consistently users interacted with the interface and how satisfaction responses were distributed across participants.

	Measures of Central Tendency			Measures of Variability			
	Mean	Median	Mode	Range	Variance	Standard Deviation	Skewness
number of clicks for Design A	161.72	161.50	159	19	21.121	4.596	-0.080
overall learner satisfaction	2.78	3.00	2	4	1.378	1.174	0.796

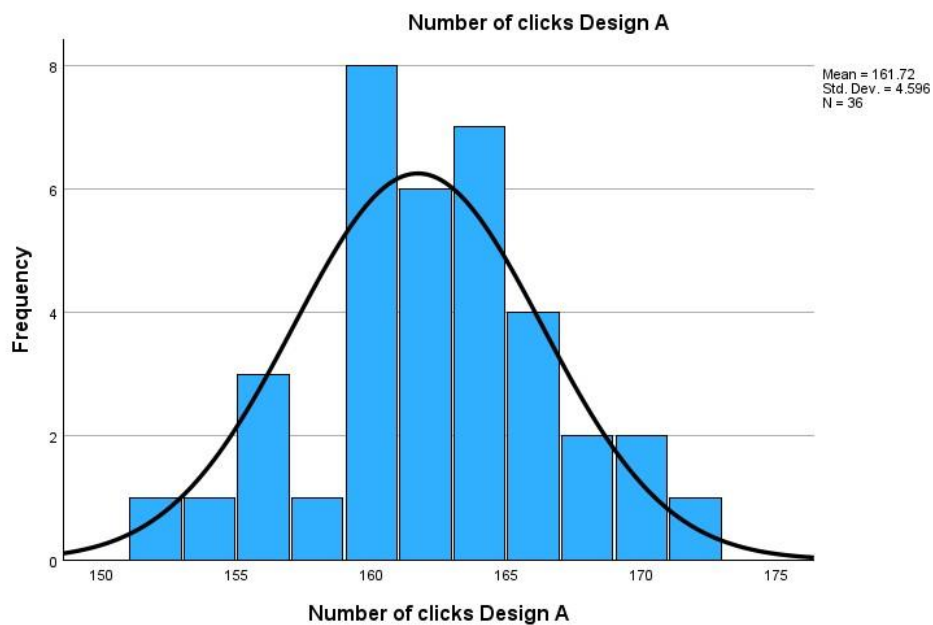
Graphical Representation

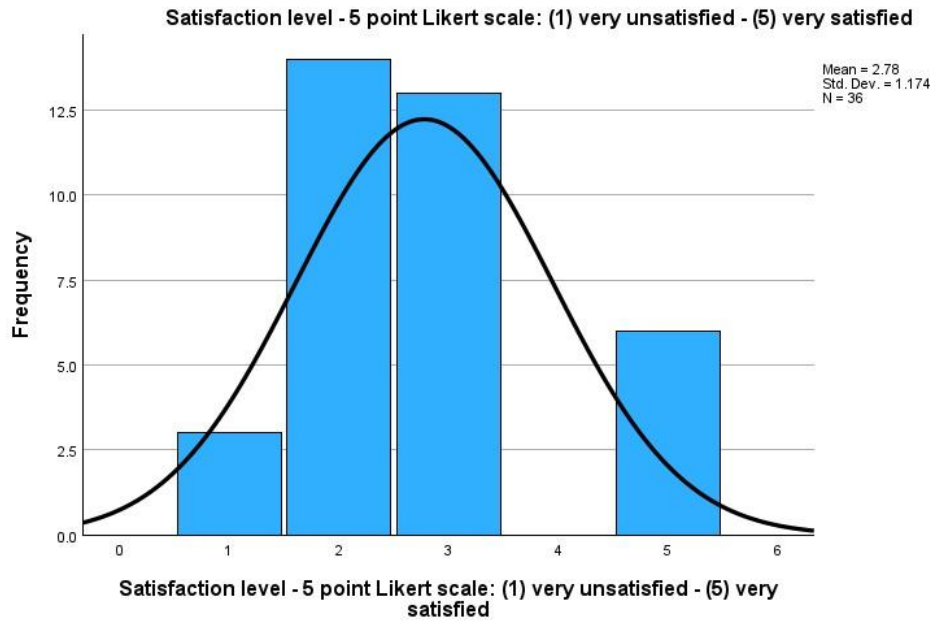
Scatter plot

A scatter plot was used to visualise the relationship between interaction effort and learner satisfaction, as it allows direct observation of the direction and strength of association between two variables.



Histogram





A histogram was used to examine the distribution of the number of clicks, as it is continuous data and supports assessment of normality.

Analysis of Descriptive Statistics

Descriptive statistics were calculated for interaction effort (number of clicks for Design A) and overall learner satisfaction for desktop users. The results show that the average number of clicks required to complete tasks using Design A was 161.72, indicating a moderate level of interaction effort. Learner satisfaction had a mean value of 2.78 on the 5-point Likert scale, suggesting that overall satisfaction tended to be neutral to slightly dissatisfied.

The variability in the number of clicks was moderate, while satisfaction scores showed a limited range of values due to the ordinal nature of the Likert scale. These results provide an overview of how interaction effort and satisfaction are distributed among desktop users prior to correlation analysis.

Analysis of Graphical Representations

Histogram Analysis

The histogram for the number of clicks for Design A appears approximately symmetric, indicating no extreme skewness and suggesting that the distribution does not deviate substantially from normality. The histogram for learner satisfaction shows clustering around the mid-scale values, which is expected for Likert-scale data and reflects the ordinal nature of the variable.

These graphical observations are consistent with the descriptive statistics and provide additional support for examining the relationship between the two variables.

Scatter Plot Analysis

The scatter plot illustrates the relationship between interaction effort and learner satisfaction. The pattern of points suggests that higher numbers of clicks are generally associated with lower satisfaction levels, while lower click counts are more often associated with higher satisfaction

ratings. This visual trend indicates a negative relationship between interaction effort and learner satisfaction.

Summary

Overall, the descriptive statistics and graphical representations suggest that increased interaction effort is associated with reduced learner satisfaction. The distributions of the variables and the observed pattern in the scatter plot support the use of correlation analysis to formally examine the strength and direction of this relationship.

2.4.4 Conducting independent sample t-test, interpret and report the results

Method recap

A Pearson correlation analysis was conducted to examine the relationship between interaction effort, measured by the number of clicks for Design A, and overall learner satisfaction for desktop users. This method was appropriate as the aim was to assess the strength and direction of association between two variables.

Descriptive Recap

The number of clicks for Design A had a mean of 161.72 (SD = 4.60), indicating a moderate level of interaction effort. Learner satisfaction had a mean score of 2.78 (SD = 1.17) on a 5point Likert scale, suggesting that satisfaction levels were generally neutral to slightly dissatisfied.

Significance Test Results

The Pearson correlation test revealed a strong negative and statistically significant relationship between the number of clicks and learner satisfaction,

$r = -0.912$, $p < .001$ (two-tailed), with $N = 36$.

This result demonstrates that higher interaction effort, reflected by a greater number of clicks, is associated with lower learner satisfaction.

Hypothesis Test Recap

- **Null Hypothesis (H_0):** There is no significant relationship between the number of clicks for Design A and overall learner satisfaction.
- **Alternative Hypothesis (H_a):** There is a significant relationship between the number of clicks for Design A and overall learner satisfaction

As the p-value is less than the significance level of 0.05, the null hypothesis is rejected. This indicates a statistically significant relationship between interaction effort and learner satisfaction for desktop users.

Reliability of the Analysis

Although learner satisfaction was measured using a 5-point Likert scale, Pearson correlation is commonly used in usability studies involving such data. The strong correlation observed, supported by graphical analysis, suggests that the results are reliable. If a non-parametric alternative were required, Spearman's rank correlation would be the appropriate test.

Pearson's Correlation results

Correlations			
		Number of clicks Design A	Satisfaction level - 5 point Likert scale: (1) very unsatisfied - (5) very satisfied
Number of clicks Design A	Pearson Correlation	1	-.912***
	Sig. (2-tailed)		<.001
	N	36	36
Satisfaction level - 5 point Likert scale: (1) very unsatisfied - (5) very satisfied	Pearson Correlation	-.912***	1
	Sig. (2-tailed)	<.001	
	N	36	36

***. Correlation at 0.001 (2-tailed)

3 Conclusions

Based on the results of the statistical analyses, the redesigned course player demonstrates clear differences in usability across layout designs. The paired-sample t-test showed that Design A enables significantly faster task completion than Design B for desktop users, indicating higher task efficiency. This suggests that the layout and interaction design of Design A better supports users in completing core tasks with less effort.

In addition, the Pearson correlation analysis revealed a strong negative relationship between interaction effort and learner satisfaction, indicating that as the number of clicks increases, overall satisfaction decreases. This finding highlights the importance of reducing unnecessary interactions to improve the user experience. Together, these results suggest that improving task efficiency by minimizing interaction effort has a direct positive impact on learner satisfaction. Overall, Design A provides better usability performance and should be preferred as the basis for further interface refinement and deployment.